UBER DATA ANALYSIS PROJECT

1. Business Problem Understanding

Urban ride-hailing platforms like Uber face multiple operational and strategic challenges:

- High **ride cancellation rates**
- **Demand-supply imbalance** (especially during peak hours)
- Variability in **driver performance and customer experience**
- Revenue leakage due to **incomplete or failed rides**

Objective

The goal of this project is to analyze Uber ride data to:

- Identify key factors affecting **ride completion and cancellations**
- Understand **customer demand patterns**
- Evaluate **driver performance**
- Detect **revenue opportunities and losses**
- Provide **data-driven recommendations** to improve efficiency and growth

2. Dataset Summary

- Rows : 150000 – Columns : 21

- Objective columns : 10
- Numeric columns : 9
- Datetime columns : 2
- Missing value columns : 13

3. Exploratory data analysis using python

e began with data preparation and cleaning in Python:

● Data Loading: Imported the dataset using pandas.

● Initial Exploration: Used df.info() to check structure and .describe() for summary statistics.

	Date	Time	Booking ID	Booking Status	Customer ID	Vehicle Type	Pickup Location	Drop Location	Avg VTAT	Avg CTAT	...	Reason for cancelling by Customer	Cancelled Rides by Driver	Driver Cancellation Reason	Incomplete Rides	Incom Re
0	2024-03-23	12:29:38	"CNR5884300"	No Driver Found	"CID1982111"	eBike	Palam Vihar	Jhilmil	NaN	NaN	...	NaN	NaN	NaN	NaN	
1	2024-11-29	18:01:39	"CNR1326809"	Incomplete	"CID4604802"	Go Sedan	Shastri Nagar	Gurgaon Sector 56	4.9	14.0	...	NaN	NaN	NaN	1.0	V Break
2	2024-08-23	08:56:10	"CNR8494506"	Completed	"CID9202816"	Auto	Khandsa	Malviya Nagar	13.4	25.8	...	NaN	NaN	NaN	NaN	
3	2024-10-21	17:17:25	"CNR8906825"	Completed	"CID2610914"	Premier Sedan	Central Secretariat	Inderlok	13.1	28.5	...	NaN	NaN	NaN	NaN	
4	2024-09-16	22:08:00	"CNR1950162"	Completed	"CID9933542"	Bike	Ghitorni Village	Khan Market	5.3	19.6	...	NaN	NaN	NaN	NaN	
5	2024-02-06	09:44:56	"CNR4096693"	Completed	"CID4670564"	Auto	AIIMS	Narsinghpur	5.1	18.1	...	NaN	NaN	NaN	NaN	

● Missing Data Handling: Checked for null values and imputed missing values

```
df.isnull().sum()
```

```
Date                                       0
Time                                       0
Booking ID                                 0
Booking Status                             0
Customer ID                                0
Vehicle Type                               0
Pickup Location                            0
Drop Location                              0
Avg VTAT                               10500
Avg CTAT                               48000
Cancelled Rides by Customer           139500
Reason for cancelling by Customer     139500
Cancelled Rides by Driver             123000
Driver Cancellation Reason            123000
Incomplete Rides                      141000
Incomplete Rides Reason               141000
Booking Value                          48000
Ride Distance                          48000
Driver Ratings                         57000
Customer Rating                        57000
Payment Method                         48000
dtype: int64
```

● Column Standardization: Renamed columns to snake case for better readability and documentation.

● Feature Engineering:

- ○ Created month and year column from date column.
- ○ Created hour column from time column and then dropped time column.
- Created Cancelled ride column

● Separated numerical and objective columns and performed EDA as per their requirement.

● Database Integration: Connected Python script to MYSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL

We performed structured analysis in MYSQL to answer key business questions:

1. Cancellation Rate by Vehicle Type

vehicle_type	total_bookings	cancelled	cancellation_rate
eBike	10557	2630	24.91
Go Sedan	27141	6863	25.29
Auto	37419	9323	24.92
Premier Sedan	18111	4516	24.94
Bike	22517	5652	25.10
Go Mini	29806	7427	24.92
Uber XL	4449	1089	24.48

2. Peak Hour Demand

hour	total_bookings
18	12397
19	11047
17	11044
16	9633
20	9630
10	9577

3. Revenue Contribution by Payment Method

payment_method	total_revenue
UPI	23345101
Cash	12895649
Uber Wallet	6200898
Credit Card	5224025
Debit Card	4180510
Not Applicable	0

4. High Value Customers

customer_id	total_spend
"CID2674107"	4987
"CID7828101"	4722
"CID2706299"	4277
"CID4843078"	4228
"CID2978596"	4220
"CID5235759"	4202
"CID5789715"	4133
"CID9539119"	4109
"CID1753183"	4088
"CID8107092"	4060
"CID6777086"	4044
"CID4542328"	4032

5. Locations with Highest Ride Distance

pickup_location	avg_distance
Green Park	18.417660332541566
Gurgaon Sector 29	17.872075242718445
Malviya Nagar	17.79966101694916
Moti Nagar	17.736600234466597
IMT Manesar	17.727345238095243
IGNOU Road	17.697701711491447
Okhla	17.66886336154776
Sikanderpur	17.65538011695904
Anand Vihar ISBT	17.652671480144413
Sultanpur	17.58930342384889

6. Driver vs Customer Cancellation Comparison

customer_cancel	driver_cancel
10500	27000

7. Top Customers per Month

customer_id	month	total_spend	rnk
"CID9081695"	1	4008	1
"CID7150286"	1	3637	2
"CID5545811"	1	3504	3
"CID6365151"	2	3878	1
"CID2933220"	2	3833	2
"CID2808747"	2	3795	3
"CID9804696"	3	4026	1
"CID2411356"	3	3921	2
"CID9330989"	3	3825	3
"CID6777086"	4	4044	1
"CID4542328"	4	4032	2
"CID9728655"	4	3851	3

8. Running Revenue Trend

date	daily_revenue	cumulative_revenue
2024-01-01	124057	124057
2024-01-02	113427	237484
2024-01-03	116982	354466
2024-01-04	117045	471511
2024-01-05	133692	605203
2024-01-06	175438	780641
2024-01-07	204168	984809
2024-01-08	112933	1097742
2024-01-09	129157	1226899
2024-01-10	126270	1353169
2024-01-11	122917	1476086
2024-01-12	110001	1586087

9. Hourly Performance Comparison

hour	avg_value	overall_avg
12	346.02583499857263	345.6237784722974
18	344.2832136807292	345.6237784722974
8	348.9857163678764	345.6237784722974
17	349.3009779065556	345.6237784722974
22	341.30123139128835	345.6237784722974
9	351.21982025746905	345.6237784722974
15	343.6609363569861	345.6237784722974
19	336.2469448719109	345.6237784722974
16	345.02231911138796	345.6237784722974
10	349.02297170303854	345.6237784722974
21	356.3623349376774	345.6237784722974
6	353.61995192307694	345.6237784722974

10. Worst Performing Routes

pickup_location	drop_location	total_rides	avg_rating	cance
Jahangirpuri	Connaught Place	Connaught Place	6666666666666	8
Satguru Ram Singh Marg	Khandsa	11	4.29090909090909	8
Akshardham	Hauz Khas	9	4.222222222222222	7
Lok Kalyan Marg	Ramesh Nagar	10	4.249999999999999	7
Vishwavidyalaya	Shastri Park	9	4.366666666666666	7
Tilak Nagar	Indraprastha	12	4.374999999999999	7
Vatika Chowk	Rithala	15	4.04	6
Cyber Hub	Ardee City	11	4.118181818181818	6
Rajiv Chowk	AIIMS	10	4.14	6
Vaishali	Maidan Garhi	9	4.155555555555555	6
Vidhan Sabha	Subhash Chowk	9	4.166666666666667	6
Jor Bagh	Rohini East	15	4.1933333333333325	6

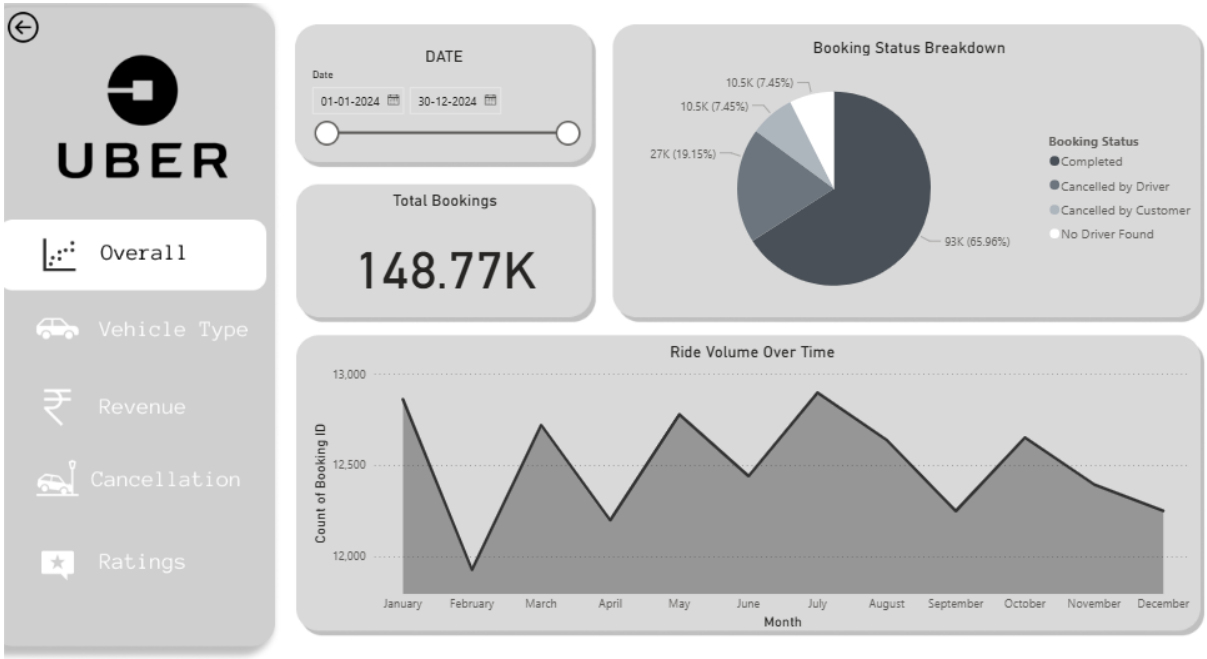
11. Demand-Supply Gap Proxy

pickup_location	total_requests	no_driver	gap_percent
Old Gurgaon	807	83	10.29
Paharganj	825	79	9.58
Vinobapuri	823	76	9.23
Rohini West	835	75	8.98
Greater Noida	866	77	8.89
Pataudi Chowk	907	80	8.82
Arjangarh	810	70	8.64
Dwarka Mor	840	72	8.57
Udyog Bhawan	838	71	8.47
Pitampura	851	72	8.46
Netaji Subhas...	830	70	8.43
Bhikaji Cama P...	821	68	8.28

12. Revenue Loss Due to Incomplete Rides

revenue_loss	actual_revenue
4585609	47260574

5. Dashboard in Power BI



UBER

Overall

Vehicle Type

Revenue

Cancellation

Ratings

01-01-2024 30-12-2024

Vehicle Type	Total Booking value	Success Booking value	Avg. Distance Travelled	Total Distance Travelled
Go Mini	10.34M	9.41M	25.99	482K
Go Sedan	9.37M	8.54M	25.98	433K
UberXL	1.53M	1.41M	25.72	72K
Premier Sedan	6.28M	5.73M	25.95	292K
Auto	12.88M	11.73M	25.99	602K
eBike/Bike	11.46M	10.44M	26.11	537K

UBER

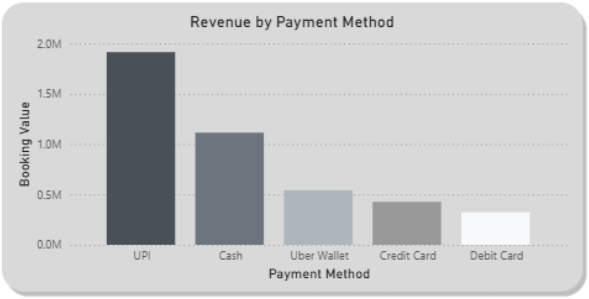
Overall

Vehicle Type

Revenue

Cancellation

Ratings

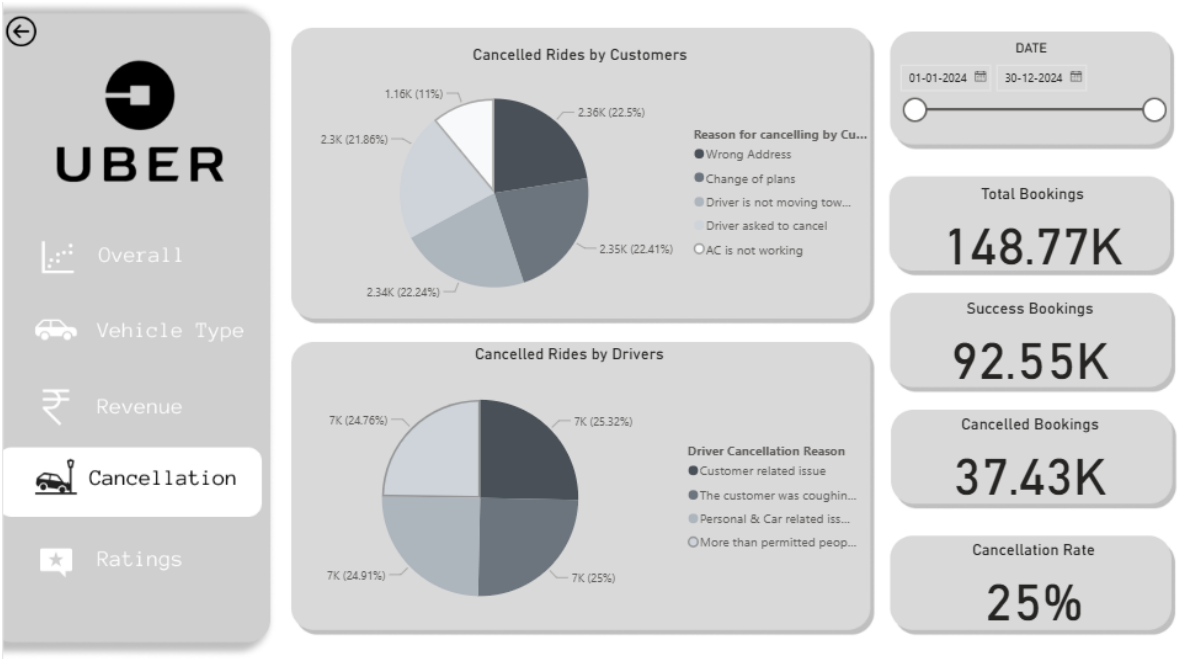


Top 5 Customers

Customer ID	Sum of Booking Value
"CID2674107"	4093
"CID6182570"	3838
"CID7154344"	3893
"CID8107092"	4060
"CID9354540"	3942
Total	19826

DATE

01-01-2024 30-12-2024



6. Business Recommendations

1. Improve Driver Supply in High-Demand Areas

- Focus on locations with:
  - High bookings
  - High “No Driver Found”

2. Optimize Peak Hour Strategy

- Introduce or refine **surge pricing**

3. Reduce Driver Cancellations

- High driver cancellations hurt reliability

4. Improve Customer Experience

- Low ratings indicate poor ride experience

5. Focus on High-Value Customers

- Identify and retain top spenders

6. Reduce Revenue Leakage

- Minimize cancellations